

Beyond Process Model Complexity: A Multi-Granular Investigation Across Time and Space Based on Eye-tracking

Appendix: List of metrics

Complexity metrics

Essential complexity

Table 1 presents a subset of the metrics that have been associated to model-related characteristics in [Mendling2008] and [Mendling2012].

Category	Description	Name/symbol	References
<i>FROM [Mendling2008]</i>			
Size	The number of nodes in the model (e.g., tasks, gateways, events).	Size, diameter (Diam)	Mendling2008, Sanchez-Gonzalez2010, Mendling2012
Density (‘Connection’ in [Mendling2012])	Relates the number of edges (possible flows) to the size of the model.	Coeff. of connectivity (Conn. Coeff.), average degree of a connector (Avg d_c), maximum degree of a connector (Max d_c)	Mendling2008, Mendling2012
Partitionability (‘Modularity’ in [Mendling2012])	Considers the relationship of subcomponents to the overall model	Separability (Π), Sequentiality (Ξ), depth (Λ), Structuredness (Φ)	Mendling2008, Figl2011, Mendling2012
Connector interplay	Considers the interactions and effects of the different connector types	Connector Heterogeneity (CH), Control Flow Complexity (CFC)	Cardoso2006, Mendling2008, Mendling2012
Cyclicity (merged in ‘Complex behavior’ in [Mendling2012])	Counts the number of nodes for which a cycle exists then provide the ratio of this number to the total number of nodes of the model.	Cyclicity (CYC)	Mendling2008, Mendling2012
Concurrency (merged in ‘Complex behavior’ in [Mendling2012])	Explores the possible concurrent paths of a model. The Token split metrics counts the control tokens associated with the control (e.g. AND or OR) designed in the model	Token split (TS)	Mendling2008, Mendling2012

Table 1 - List of metrics addressing essential complexity.

Accidental complexity

Table 2 summarizes a list of metrics provided by [Bernstein2015] and [Burattin2017] (detailed formulas can be found in the cited studies) with name and the description of each feature category:

Categories	Description	Name / Symbol	Reference (support the features)
<i>From [Bernstein2015]</i>			
Edges style	A measure of the style of the edges as the ratio of simple (default) or 'broken' (with breaking points) edges to the total number of edges.	%simpleEdges (%sE), %brokenEdges (%bE)	[Purchase1997], [Schrepfer2009], [Effinger2010]
Crossing edges	Ratio of the number of crossing edges to the total number of edges	%totalCross (%tC)	[Purchase1997], [Schrepfer2009], [Effinger2010]
Angles	Ratio of the number of orthogonal segments to the total number of segments. <i>Orthogonal segments are parts of edges which are aligned with a grid layout of the model.</i>	%orthogonalSegments (%oS)	[Purchase1997], [Effinger2010]
Symmetry in blocks*	Symmetry of the elements' arrangement inside a block of the model.	%symmetricalPatterns (%sP)	<i>(See note on symmetry in blocks in the table caption)</i>
<i>From [Burattin2017]</i>			
Flow consistency	Measure the extent to which the model maintains its general flow direction.	Metric based on behavioral profiles (M-BP)	[Effinger2010]

Table 2 - List of metrics addressing accidental complexity proposed by [Bernstein2015] and [Burattin2017].

(*) The authors in [Bernstein2015] propose the concept of symmetry in blocks as a category of visual features that affect positively the reading/understanding of models, but did not provide any quantification.

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